

Laptop Price Prediction Using Machine Learning Algorithms

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Abstract

Laptop price prediction is an important application in e-commerce and market analysis, helping both buyers and sellers make informed decisions. The price of a laptop depends on multiple hardware and software specifications such as memory, processor type, graphics card, storage type, and display features, which makes manual estimation challenging.

This study presents a machine learning-based system for predicting laptop prices. The dataset is preprocessed using standard techniques including data cleaning, label encoding, feature extraction, and feature scaling to improve model performance.

Several machine learning algorithms are implemented and evaluated to identify the most accurate model for prediction. Model performance is analyzed using accuracy metrics and visualization techniques.

The results show that the Support Vector Machine achieves the highest accuracy of approximately 95%, outperforming other models. The proposed approach demonstrates that proper preprocessing and model evaluation significantly improve prediction performance, making it suitable for real-world applications.

Keywords: Machine Learning, Laptop Price Prediction, Support Vector Machine, XGBoost, Data Preprocessing, Artificial Intelligence

1 Introduction

Machine Learning is one of the most important branches of Artificial Intelligence that enables computers to learn patterns from data and make intelligent predictions. In recent years, Machine Learning techniques have been widely applied in different domains including healthcare, finance, recommendation systems, and e-commerce.

Laptop price prediction is an important problem because laptop prices depend on multiple specifications such as RAM, processor, GPU, storage type, operating system, and screen

features. Manual estimation of laptop prices is difficult because many hardware and software factors influence the final market price.

The main objective of this research is to develop an intelligent laptop price prediction system using Machine Learning algorithms such as Random Forest [1], XGBoost [2], and Support Vector Machine [3] which are widely used in predictive modeling due to their strong performance on structured datasets. Multiple supervised learning models are implemented and compared to identify the best-performing algorithm for accurate prediction.

The proposed system improves previous work by adding multiple Machine Learning algorithms, preprocessing techniques, visualization methods, parameter tuning, and detailed performance analysis.

2 Literature Review

Several researchers have applied Machine Learning techniques for product price prediction and market analysis. Previous studies mainly focused on basic prediction systems using limited Machine Learning algorithms.

The base research paper used in this study [4] focuses on comparative analysis of machine learning models for laptop price prediction using supervised learning techniques.

Different Machine Learning algorithms such as Decision Tree, Random Forest, and Support Vector Machine have been widely used in prediction systems because of their ability to analyze complex datasets. Researchers have also applied ensemble learning techniques including Gradient Boosting and XGBoost to improve prediction performance.

However, many previous studies lacked advanced preprocessing techniques, detailed comparison analysis, visualization, and hyperparameter tuning. The proposed research addresses these limitations by implementing multiple Machine Learning models, advanced preprocessing, visualization techniques, and performance evaluation.

3 Problem Statement

Laptop prices are influenced by multiple hardware and software specifications including RAM, processor, GPU, storage type, operating system, and screen features. Traditional methods of price estimation are often inaccurate because many factors affect the final price.

The main problem addressed in this research is:

“How to accurately predict laptop prices using Machine Learning algorithms?”

4 Objectives

The main objectives of this research are:

- To develop a Machine Learning-based laptop price prediction system.
- To preprocess and clean the dataset for accurate prediction.
- To apply and compare multiple Machine Learning algorithms.
- To improve prediction accuracy using preprocessing and parameter tuning.
- To identify the best-performing Machine Learning model.
- To visualize dataset patterns and model performance using graphs and charts.

5 Proposed Methodology

5.1 Dataset Collection

The dataset used in this project contains laptop specifications and corresponding prices. The dataset was obtained from Kaggle [5] and includes several important features such as:

- Company Name
- Laptop Type
- RAM
- CPU
- GPU
- Storage
- Weight
- Operating System
- Price

5.2 Data Preprocessing

Data preprocessing was performed to improve data quality and prediction accuracy. The following preprocessing techniques were applied:

- Removing unnecessary columns
- Handling missing values
- Label encoding
- Feature extraction
- Data scaling

5.3 Exploratory Data Analysis

Different visualization techniques were used to understand the dataset and identify important patterns.

The following visualizations were used:

- Histogram Plot
- Box Plot

- Scatter Plot
- Heatmap
- Bar Chart
- PCA Scatter Plot
- Confusion Matrix Heatmap

5.4 Machine Learning Models

Multiple Machine Learning algorithms were implemented and compared in this research.

5.4.1 Logistic Regression

Logistic Regression was used as a baseline Machine Learning model for prediction.

5.4.2 Decision Tree

Decision Tree was used because of its simplicity and ability to handle non-linear data.

5.4.3 Random Forest

Random Forest combines multiple decision trees to improve prediction accuracy and reduce overfitting.

5.4.4 Gradient Boosting

Gradient Boosting improves prediction performance by combining weak learners sequentially.

5.4.5 AdaBoost

AdaBoost focuses on correcting prediction errors by assigning higher importance to misclassified samples.

5.4.6 XGBoost

XGBoost is an advanced boosting algorithm that provides efficient and accurate prediction performance.

5.4.7 K-Nearest Neighbors

KNN predicts values based on similarity between neighboring data points.

5.4.8 Support Vector Machine

Support Vector Machine [3] achieved the highest prediction accuracy in this research project due to its strong ability to handle high-dimensional data.

6 Model Training and Evaluation

6.1 Train-Test Split

The dataset was divided into training and testing datasets. Approximately 80% of the data was used for training while 20% was used for testing.

6.2 Hyperparameter Tuning

Parameter tuning techniques were applied to improve model performance.

The following methods were used:

- GridSearchCV
- RandomizedSearchCV

6.3 Evaluation Metrics

Different evaluation metrics were used to compare model performance.

- Accuracy Score
- Confusion Matrix
- Classification Report
- Performance Comparison

7 System Architecture

The proposed system follows the following workflow:

1. Dataset Collection
2. Data Preprocessing
3. Feature Engineering
4. Model Training
5. Model Evaluation
6. Best Model Selection
7. Laptop Price Prediction

8 Results and Discussion

Different Machine Learning models were trained and evaluated for laptop price prediction.

Experimental results showed that Support Vector Machine achieved the highest prediction accuracy among all implemented models.

9 Comparative Analysis

The proposed system improved previous research work by adding advanced preprocessing

Table 1 Accuracy Comparison of Machine Learning Models

Model	Accuracy
Random Forest	84.26%
Logistic Regression	68.53%
Decision Tree	74.57%
Gradient Boosting	84.78%
XGBoost	85.37%
Support Vector Machine	95.35%

techniques, multiple Machine Learning models, visualization methods, and parameter tuning.

Table 2 Comparison Between Base Research Paper and Proposed System

Feature	Base Paper	Proposed System
Machine Learning Models	Limited Models	Multiple Models
Visualization	Basic	Advanced Visualization Applied
Parameter Tuning	Limited	
Performance Evaluation	Basic	Detailed Comparison
Best Model Selection	Not Clear	Clearly Identified

10 Deep Learning Approach

A Deep Learning approach was also explored for experimental comparison purposes using TensorFlow and Keras. Although traditional Machine Learning models performed better for this dataset, this extension was included to analyze the potential of neural network-based approaches for future scalability.

11 Conclusion

This research successfully developed a Machine Learning-based laptop price prediction system using multiple supervised learning algorithms.

Different preprocessing techniques, visualization methods, parameter tuning techniques, and Machine Learning models were implemented and compared. Experimental results

demonstrated that Support Vector Machine achieved the highest accuracy of approximately 95%.

The proposed system provides better performance, preprocessing, visualization, and evaluation compared to previous research approaches.

12 Future Work

Future improvements for this research may include:

- Using larger datasets
- Implementing real-time price prediction systems
- Deploying the model as a web application
- Applying advanced Deep Learning architectures
- Using hybrid Machine Learning models

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Declarations

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Conflict of Interest

The author declares no conflict of interest.

Data Availability

The dataset used in this research was obtained from Kaggle.

Code Availability

The implementation code and Jupyter Notebook are publicly available on GitHub.

References

- [1] Breiman, L.: Random forests. Machine Learning **45**(1), 5–32 (2001) <https://doi.org/10.1023/A:1010933404324>
- [2] Chen, T., Guestrin, C.: Xgboost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (2016). <https://doi.org/10.1145/2939672.2939785>

- [3] Cortes, C., Vapnik, V.: Support-vector networks. Machine Learning (1995) <https://doi.org/10.1007/BF00994018>
- [4] Kolla, V.R.K.: Forecasting Laptop Prices: A Comparative Study of Machine Learning Algorithms for Predictive Modeling. SSRN Electronic Journal, Accessed: May 2026 (2023). https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4413726
- [5] Rahul, V.: Laptop Price Dataset (Most Detailed). Accessed: May 2026. <https://www.kaggle.com/datasets/vashisthrahul13/laptop-price-dataset-most-detailed>